

# **Data Mining:**

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## **Concepts and Techniques**

**(3<sup>rd</sup> ed.)**

### **— Chapter 7 —**

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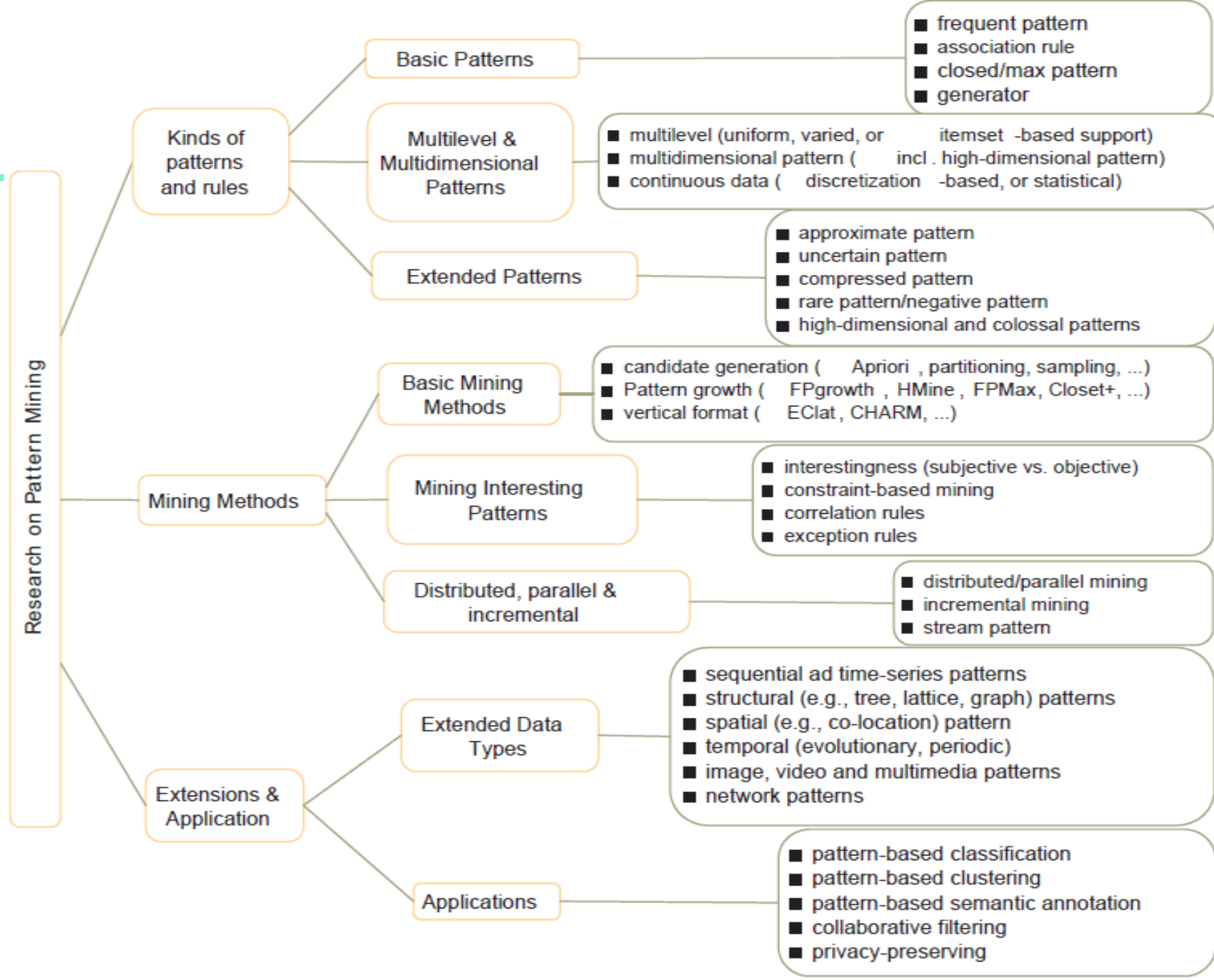
# Chapter 7 : Advanced Frequent Pattern Mining

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- ⑩ Pattern Mining: A Road Map 
- ⑩ Pattern Mining in Multi-Level, Multi-Dimensional Space
- ⑩ Constraint-Based Frequent Pattern Mining
- ⑩ Mining High-Dimensional Data and Colossal Patterns
- ⑩ Mining Compressed or Approximate Patterns
- ⑩ Pattern Exploration and Application
- ⑩ Summary



# Research on Pattern Mining: A Road Map



# Chapter 7 : Advanced Frequent Pattern Mining

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  - Mining Multi-Level Association 
  - Mining Multi-Dimensional Association
  - Mining Quantitative Association Rules
  - Mining Rare Patterns and Negative Patterns
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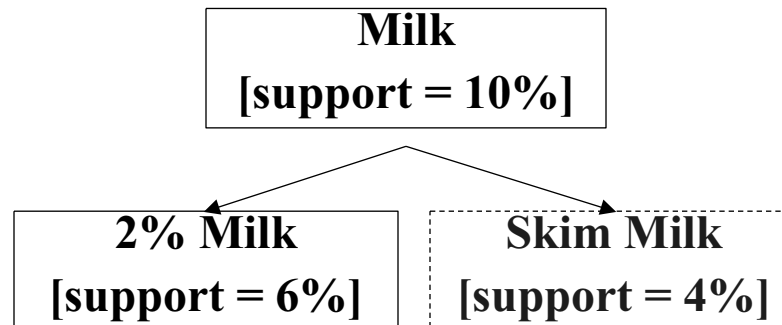
# Mining Multiple-Level Association Rules

- Items often form hierarchies
- Flexible support settings
  - Items at the lower level are expected to have lower support
- Exploration of *shared* multi-level mining (Agrawal & Srikant@VLB'95, Han & Fu@VLDB'95)

uniform support

Level 1  
min\_sup = 5%

Level 2  
min\_sup = 5%



reduced support

Level 1  
min\_sup = 5%

Level 2  
min\_sup = 3%

# Multi-level Association: Flexible Support and Redundancy filtering

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- Flexible min-support thresholds: Some items are more valuable but less frequent
  - Use non-uniform, group-based min-support
  - E.g., {diamond, watch, camera}: 0.05%; {bread, milk}: 5%; ...
- Redundancy Filtering: Some rules may be redundant due to “ancestor” relationships between items
  - milk  $\Rightarrow$  wheat bread [support = 8%, confidence = 70%]
  - 2% milk  $\Rightarrow$  wheat bread [support = 2%, confidence = 72%]

The first rule is an ancestor of the second rule

- A rule is *redundant* if its support is close to the “expected” value, based on the rule’s ancestor

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# Mining Multi-Dimensional Association

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- Single-dimensional rules:

$\text{buys}(X, \text{"milk"}) \Rightarrow \text{buys}(X, \text{"bread"})$

- Multi-dimensional rules:  $\geq 2$  dimensions or predicates

- Inter-dimension assoc. rules (*no repeated predicates*)

$\text{age}(X, \text{"19-25"}) \wedge \text{occupation}(X, \text{"student"}) \Rightarrow \text{buys}(X, \text{"coke"})$

- hybrid-dimension assoc. rules (*repeated predicates*)

$\text{age}(X, \text{"19-25"}) \wedge \text{buys}(X, \text{"popcorn"}) \Rightarrow \text{buys}(X, \text{"coke"})$

- Categorical Attributes: finite number of possible values, no ordering among values—data cube approach

- Quantitative Attributes: Numeric, implicit ordering among values—discretization, clustering, and gradient approaches

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# Mining Quantitative Associations

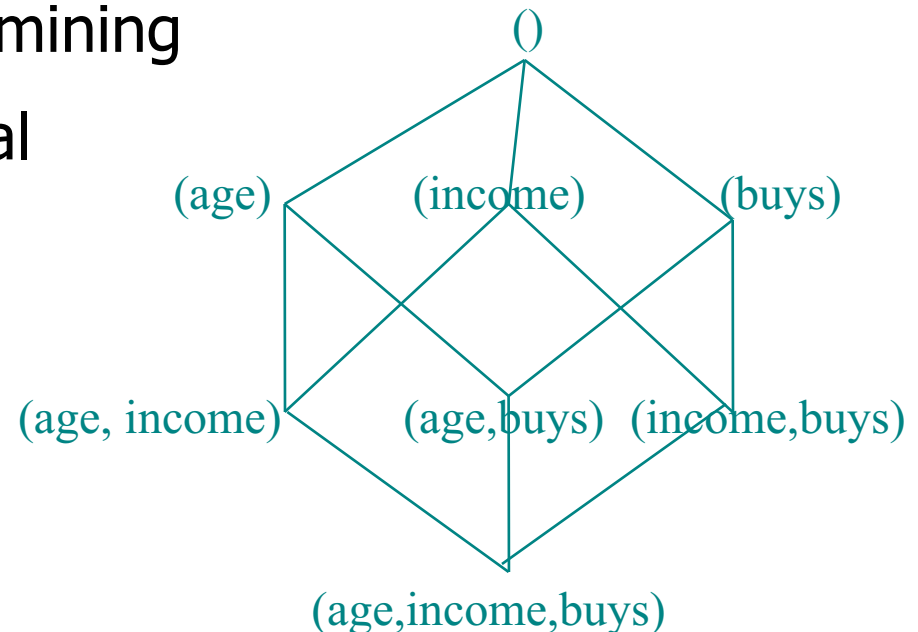
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Techniques can be categorized by how numerical attributes, such as **age** or **salary** are treated

1. Static discretization based on predefined concept hierarchies (data cube methods)
2. Dynamic discretization based on data distribution (quantitative rules, e.g., Agrawal & Srikant@SIGMOD96)
3. Clustering: Distance-based association (e.g., Yang & Miller@SIGMOD97)
  - ⑩ One dimensional clustering then association
4. Deviation: (such as Aumann and Lindell@KDD99)
  - Sex = female => Wage: mean=\$7/hr (overall mean = \$9)

# Static Discretization of Quantitative Attributes

- Discretized prior to mining using concept hierarchy.
- Numeric values are replaced by ranges
- In relational database, finding all frequent  $k$ -predicate sets will require  $k$  or  $k+1$  table scans
- Data cube is well suited for mining
- The cells of an  $n$ -dimensional cuboid correspond to the predicate sets
- Mining from data cubes can be much faster



# Quantitative Association Rules Based on Statistical Inference Theory [Aumann and Lindell@DMKD'03]

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- Finding extraordinary and therefore interesting phenomena, e.g.,  
(Sex = female)  $\Rightarrow$  Wage: mean=\$7/hr (overall mean = \$9)
  - LHS: a subset of the population
  - RHS: an extraordinary behavior of this subset
- The rule is accepted only if a statistical test (e.g., Z-test) confirms the inference with high confidence
- Subrule: highlights the extraordinary behavior of a subset of the pop. of the super rule
  - E.g., (Sex = female)  $\wedge$  (South = yes)  $\Rightarrow$  mean wage = \$6.3/hr
- Two forms of rules
  - Categorical  $\Rightarrow$  quantitative rules, or Quantitative  $\Rightarrow$  quantitative rules
  - E.g., Education in [14-18] (yrs)  $\Rightarrow$  mean wage = \$11.64/hr
- Open problem: Efficient methods for LHS containing two or more quantitative attributes

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# Negative and Rare Patterns

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- Rare patterns: Very low support but interesting
  - E.g., buying Rolex watches
  - Mining: Setting individual-based or special group-based support threshold for valuable items
- Negative patterns
  - Since it is unlikely that one buys Ford Expedition (an SUV car) and Toyota Prius (a hybrid car) together, Ford Expedition and Toyota Prius are likely negatively correlated patterns
- Negatively correlated patterns that are infrequent tend to be more interesting than those that are frequent

# Defining Negative Correlated Patterns (I)

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- Definition 1 (support-based)
  - If itemsets X and Y are both frequent but rarely occur together, i.e.,
$$\text{sup}(X \cup Y) < \text{sup}(X) * \text{sup}(Y)$$
  - Then X and Y are negatively correlated
- Problem: A store sold two needle 100 packages A and B, only one transaction containing both A and B.
  - When there are in total 200 transactions, we have
$$s(A \cup B) = 0.005, s(A) * s(B) = 0.25, s(A \cup B) < s(A) * s(B)$$
  - When there are  $10^5$  transactions, we have
$$s(A \cup B) = 1/10^5, s(A) * s(B) = 1/10^3 * 1/10^3, s(A \cup B) > s(A) * s(B)$$
  - Where is the problem? —Null transactions, i.e., the support-based definition is not null-invariant!

# Defining Negative Correlated Patterns (II)

- Definition 2 (negative itemset-based)

- $X$  is a *negative itemset* if (1)  $X = \bar{A} \cup B$ , where  $B$  is a set of positive items, and  $\bar{A}$  is a set of negative items,  $|\bar{A}| \geq 1$ , and (2)  $s(X) \geq \mu$
- Itemsets  $X$  is negatively correlated, if

$$s(X) < \prod_{i=1}^k s(x_i), \text{ where } x_i \in X, \text{ and } s(x_i) \text{ is the support of } x_i$$

- This definition suffers a similar null-invariant problem
- Definition 3 (Kulczynski measure-based) If itemsets  $X$  and  $Y$  are frequent, but  $(P(X|Y) + P(Y|X))/2 < \epsilon$ , where  $\epsilon$  is a negative pattern threshold, then  $X$  and  $Y$  are negatively correlated.
- Ex. For the same needle package problem, when no matter there are 200 or  $10^5$  transactions, if  $\epsilon = 0.01$ , we have

$$(P(A|B) + P(B|A))/2 = (0.01 + 0.01)/2 < \epsilon$$

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# Constraint-based (Query-Directed) Mining

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- Finding **all** the patterns in a database **autonomously**? — unrealistic!
  - The patterns could be too many but not focused!
- Data mining should be an **interactive** process
  - User directs what to be mined using a **data mining query language** (or a graphical user interface)
- Constraint-based mining
  - User flexibility: provides **constraints** on what to be mined
  - Optimization: explores such constraints for efficient mining — **constraint-based mining**: constraint-pushing, similar to push selection first in DB query processing
  - Note: still find all the answers satisfying constraints, not finding some answers in “heuristic search”

# Constraints in Data Mining

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- Knowledge type constraint:
  - classification, association, etc.
- Data constraint — using SQL-like queries
  - find product pairs sold together in stores in Chicago this year
- Dimension/level constraint
  - in relevance to region, price, brand, customer category
- Rule (or pattern) constraint
  - small sales (price < \$10) triggers big sales (sum > \$200)
- Interestingness constraint
  - strong rules: min\_support  $\geq 3\%$ , min\_confidence  $\geq 60\%$

# Meta-Rule Guided Mining

- Meta-rule can be in the rule form with partially instantiated predicates and constants

$$P_1(X, Y) \wedge P_2(X, W) \Rightarrow \text{buys}(X, \text{"iPad"})$$

- The resulting rule derived can be

$$\text{age}(X, \text{"15-25"}) \wedge \text{profession}(X, \text{"student"}) \Rightarrow \text{buys}(X, \text{"iPad"})$$

- In general, it can be in the form of

$$P_1 \wedge P_2 \wedge \dots \wedge P_l \Rightarrow Q_1 \wedge Q_2 \wedge \dots \wedge Q_r$$

- Method to find meta-rules
  - Find frequent (l+r) predicates (based on min-support threshold)
  - Push constants deeply when possible into the mining process (see the remaining discussions on constraint-push techniques)
  - Use confidence, correlation, and other measures when possible



# Constraint-Based Frequent Pattern Mining

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- Pattern space pruning constraints
  - **Anti-monotonic**: If constraint  $c$  is violated, its further mining can be terminated
  - **Monotonic**: If  $c$  is satisfied, no need to check  $c$  again
  - **Succinct**:  $c$  must be satisfied, so one can start with the data sets satisfying  $c$
  - **Convertible**:  $c$  is not monotonic nor anti-monotonic, but it can be converted into it if items in the transaction can be properly ordered
- Data space pruning constraint
  - **Data succinct**: Data space can be pruned at the initial pattern mining process
  - **Data anti-monotonic**: If a transaction  $t$  does not satisfy  $c$ ,  $t$  can be pruned from its further mining

# Pattern Space Pruning with Anti-Monotonicity Constraints

- A constraint  $C$  is *anti-monotone* if the super pattern satisfies  $C$ , all of its sub-patterns do so too
- In other words, *anti-monotonicity*: If an itemset  $S$  **violates** the constraint, so does any of its superset
- Ex. 1.  $\text{sum}(S.\text{price}) \leq v$  is **anti-monotone**
- Ex. 2.  $\text{range}(S.\text{profit}) \leq 15$  is **anti-monotone**
  - Itemset  $ab$  violates  $C$
  - So does every superset of  $ab$
- Ex. 3.  $\text{sum}(S.\text{Price}) \geq v$  is **not anti-monotone**
- Ex. 4. *support count* is anti-monotone: core property used in Apriori

TDB (min\_sup=2)

| TID | Transaction      |
|-----|------------------|
| 10  | a, b, c, d, f    |
| 20  | b, c, d, f, g, h |
| 30  | a, c, d, e, f    |
| 40  | c, e, f, g       |

| Item | Profit |
|------|--------|
| a    | 40     |
| b    | 0      |
| c    | -20    |
| d    | 10     |
| e    | -30    |
| f    | 30     |
| g    | 20     |
| h    | -10    |

# Pattern Space Pruning with Monotonicity Constraints

- A constraint  $C$  is *monotone* if the pattern satisfies  $C$ , we do not need to check  $C$  in subsequent mining
- Alternatively, monotonicity: *If an itemset  $S$  **satisfies** the constraint, so does any of its superset*
- Ex. 1.  $\text{sum}(S.\text{Price}) \geq v$  is **monotone**
- Ex. 2.  $\text{min}(S.\text{Price}) \leq v$  is **monotone**
- Ex. 3.  $C: \text{range}(S.\text{profit}) \geq 15$ 
  - Itemset  $ab$  satisfies  $C$
  - So does every superset of  $ab$

TDB (min\_sup=2)

| TID | Transaction      |
|-----|------------------|
| 10  | a, b, c, d, f    |
| 20  | b, c, d, f, g, h |
| 30  | a, c, d, e, f    |
| 40  | c, e, f, g       |

| Item | Profit |
|------|--------|
| a    | 40     |
| b    | 0      |
| c    | -20    |
| d    | 10     |
| e    | -30    |
| f    | 30     |
| g    | 20     |
| h    | -10    |

# Data Space Pruning with Data Anti-monotonicity

- A constraint  $c$  is *data anti-monotone* if for a pattern  $p$  cannot satisfy a transaction  $t$  under  $c$ ,  $p$ 's superset cannot satisfy  $t$  under  $c$  either
- The key for data anti-monotone is *recursive data reduction*
- Ex. 1.  $\text{sum}(S.\text{Price}) \geq v$  is data anti-monotone
- Ex. 2.  $\text{min}(S.\text{Price}) \leq v$  is data anti-monotone
- Ex. 3. C:  $\text{range}(S.\text{profit}) \geq 25$  is data anti-monotone
  - Itemset  $\{b, c\}$ 's projected DB:
    - T10':  $\{d, f, h\}$ , T20':  $\{d, f, g, h\}$ , T30':  $\{d, f, g\}$
  - since C cannot satisfy T10', T10' can be pruned

TDB (min\_sup=2)

| TID | Transaction      |
|-----|------------------|
| 10  | a, b, c, d, f, h |
| 20  | b, c, d, f, g, h |
| 30  | b, c, d, f, g    |
| 40  | c, e, f, g       |

| Item | Profit |
|------|--------|
| a    | 40     |
| b    | 0      |
| c    | -20    |
| d    | -15    |
| e    | -30    |
| f    | -10    |
| g    | 20     |
| h    | -5     |

# Pattern Space Pruning with Succinctness

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- Succinctness:
  - Given  $A_1$ , the set of items satisfying a succinctness constraint  $C$ , then any set  $S$  satisfying  $C$  is based on  $A_1$ , i.e.,  $S$  contains a subset belonging to  $A_1$
  - Idea: Without looking at the transaction database, whether an itemset  $S$  satisfies constraint  $C$  can be determined based on the selection of items
  - $\min(S.Price) \leq v$  is succinct
  - $\sum(S.Price) \geq v$  is not succinct
- Optimization: If  $C$  is succinct,  $C$  is pre-counting pushable

# Naïve Algorithm: Apriori + Constraint

Database D

| TID | Items   |
|-----|---------|
| 100 | 1 3 4   |
| 200 | 2 3 5   |
| 300 | 1 2 3 5 |
| 400 | 2 5     |

Scan D

$C_1$

| itemset | sup. |
|---------|------|
| {1}     | 2    |
| {2}     | 3    |
| {3}     | 3    |
| {4}     | 1    |
| {5}     | 3    |

$L_1$

| itemset        | sup.         |
|----------------|--------------|
| {1}            | 2            |
| {2}            | 3            |
| {3}            | 3            |
| <del>{5}</del> | <del>3</del> |

$C_2$

| itemset | sup |
|---------|-----|
| {1 2}   | 1   |
| {1 3}   | 2   |
| {1 5}   | 1   |
| {2 3}   | 2   |
| {2 5}   | 3   |
| {3 5}   | 2   |

Scan D

$C_2$

| itemset |
|---------|
| {1 2}   |
| {1 3}   |
| {1 5}   |
| {2 3}   |
| {2 5}   |
| {3 5}   |

$L_2$

| itemset          | sup          |
|------------------|--------------|
| {1 3}            | 2            |
| <del>{2 3}</del> | <del>2</del> |
| <del>{2 5}</del> | <del>3</del> |
| <del>{3 5}</del> | <del>2</del> |

$C_3$

| itemset |
|---------|
| {2 3 5} |

Scan D

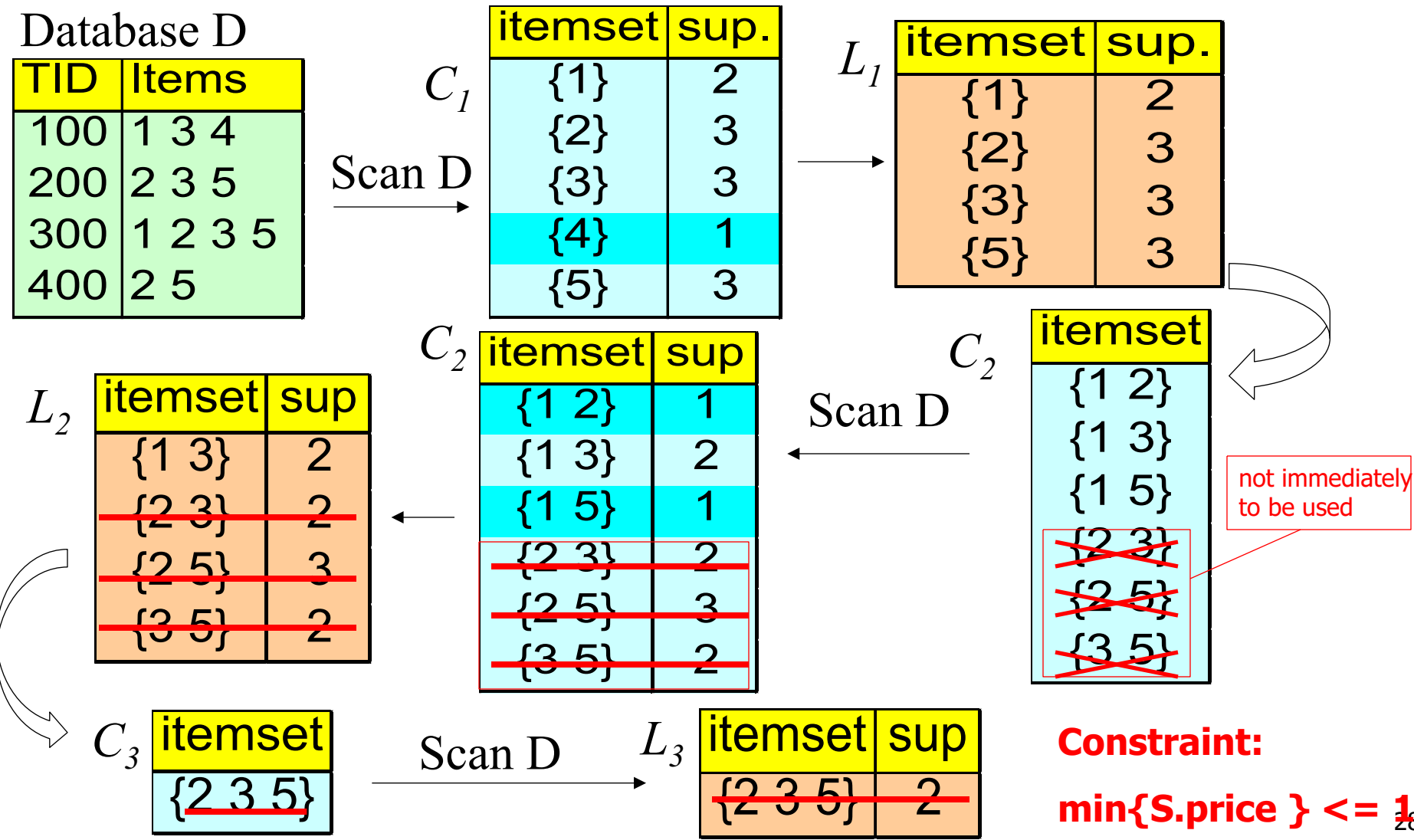
$L_3$

| itemset            | sup          |
|--------------------|--------------|
| <del>{2 3 5}</del> | <del>2</del> |

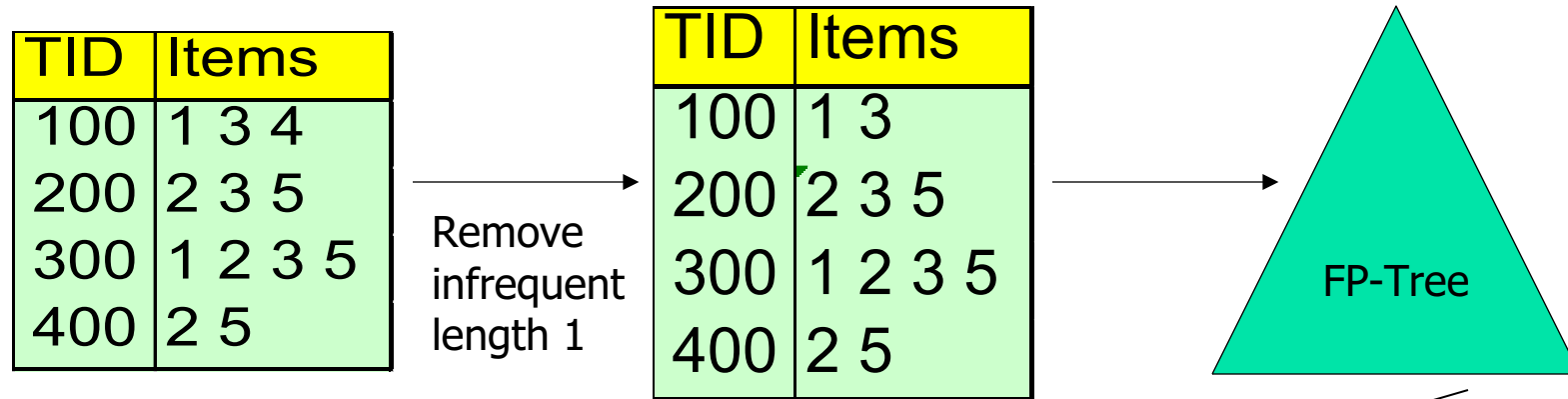
**Constraint:**

**Sum{S.price} < 5** <sub>27</sub>

# Constrained Apriori : Push a Succinct Constraint Deep



# Constrained FP-Growth: Push a Succinct Constraint Deep



1-Projected DB

| TID | Items |
|-----|-------|
| 100 | 3 4   |
| 300 | 2 3 5 |

No Need to project on 2, 3, or 5

**Constraint:**

**$\min\{S.\text{price}\} \leq 1$**

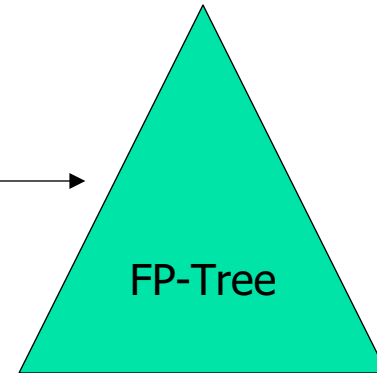


# Constrained FP-Growth: Push a Data Anti-monotonic Constraint Deep

Remove from data

| TID            | Items            |
|----------------|------------------|
| 100            | 1 3 4            |
| <del>200</del> | <del>2 3 5</del> |
| 300            | 1 2 3 5          |
| <del>400</del> | <del>2 5</del>   |

| TID | Items |
|-----|-------|
| 100 | 1 3   |
| 300 | 1 3   |



Single branch, we are done

**Constraint:**

**$\min\{S.\text{price}\} \leq 1$**

# Constrained FP-Growth: Push a Data Anti-monotonic Constraint Deep

| TID | Transaction      |
|-----|------------------|
| 10  | a, b, c, d, f, h |
| 20  | b, c, d, f, g, h |
| 30  | b, c, d, f, g    |
| 40  | a, c, e, f, g    |

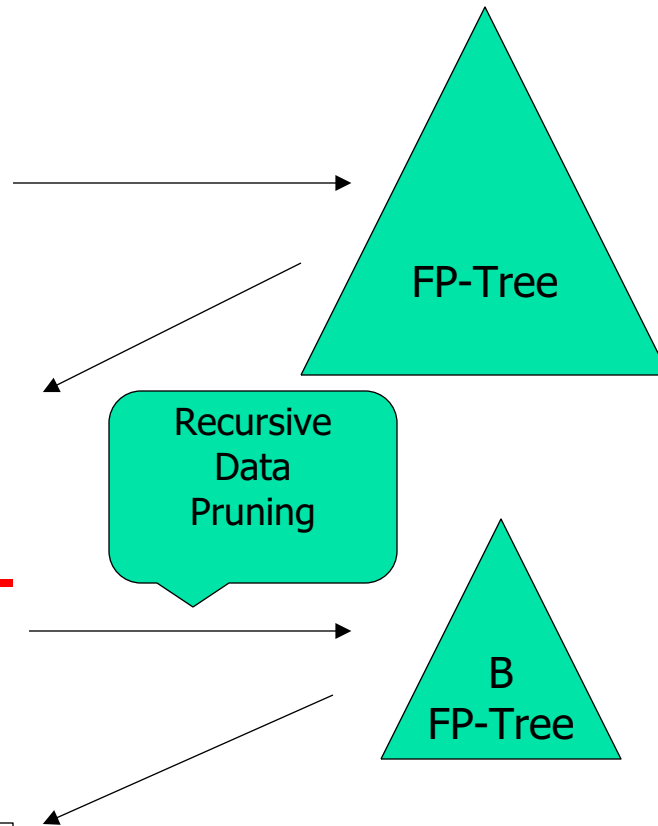
| TID | Transaction      |
|-----|------------------|
| 10  | a, b, c, d, f, h |
| 20  | b, c, d, f, g, h |
| 30  | b, c, d, f, g    |
| 40  | a, c, e, f, g    |

B-Projected DB

| TID           | Transaction              |
|---------------|--------------------------|
| <del>10</del> | <del>a, c, d, f, h</del> |
| 20            | c, d, f, g, <del>h</del> |
| 30            | c, d, f, g               |

Single branch:

bcdfg: 2



| Item | Profit |
|------|--------|
| a    | 40     |
| b    | 0      |
| c    | -20    |
| d    | -15    |
| e    | -30    |
| f    | -10    |
| g    | 20     |
| h    | -5     |

**Constraint:**  
 $\text{range}\{S.\text{price}\} > 25$   
 $\text{min\_sup} \geq 2$

# Convertible Constraints: Ordering Data in Transactions

- Convert tough constraints into anti-monotone or monotone by properly ordering items
- Examine C:  $\text{avg}(S.\text{profit}) \geq 25$ 
  - Order items in value-descending order
    - $\langle a, f, g, d, b, h, c, e \rangle$
  - If an itemset  $afb$  violates C
    - So does  $afbh, afb^*$
    - It becomes **anti-monotone!**

TDB (min\_sup=2)

| TID | Transaction      |
|-----|------------------|
| 10  | a, b, c, d, f    |
| 20  | b, c, d, f, g, h |
| 30  | a, c, d, e, f    |
| 40  | c, e, f, g       |

| Item | Profit |
|------|--------|
| a    | 40     |
| b    | 0      |
| c    | -20    |
| d    | 10     |
| e    | -30    |
| f    | 30     |
| g    | 20     |
| h    | -10    |

# Strongly Convertible Constraints

- $\text{avg}(X) \geq 25$  is convertible anti-monotone w.r.t. item **value descending** order  $R$ :  $\langle a, f, g, d, b, h, c, e \rangle$ 
  - If an itemset  $af$  violates a constraint  $C$ , so does every itemset with  $af$  as prefix, such as  $afd$
- $\text{avg}(X) \geq 25$  is convertible monotone w.r.t. item **value ascending** order  $R^{-1}$ :  $\langle e, c, h, b, d, g, f, a \rangle$ 
  - If an itemset  $d$  satisfies a constraint  $C$ , so does itemsets  $df$  and  $dfa$ , which having  $d$  as a prefix
- Thus,  $\text{avg}(X) \geq 25$  is **strongly convertible**

| Item | Profit |
|------|--------|
| a    | 40     |
| b    | 0      |
| c    | -20    |
| d    | 10     |
| e    | -30    |
| f    | 30     |
| g    | 20     |
| h    | -10    |

# Can Apriori Handle Convertible Constraints?

- A convertible, not monotone nor anti-monotone nor succinct constraint cannot be pushed deep into the an Apriori mining algorithm
  - Within the level wise framework, no direct pruning based on the constraint can be made
  - Itemset  $df$  violates constraint  $C: \text{avg}(X) \geq 25$
  - Since  $adf$  satisfies  $C$ , Apriori needs  $df$  to assemble  $adf$ ,  $df$  cannot be pruned
- But it can be pushed into frequent-pattern growth framework!

| Item | Value |
|------|-------|
| a    | 40    |
| b    | 0     |
| c    | -20   |
| d    | 10    |
| e    | -30   |
| f    | 30    |
| g    | 20    |
| h    | -10   |

# Pattern Space Pruning w. Convertible Constraints

- C:  $\text{avg}(X) \geq 25$ ,  $\text{min\_sup}=2$
- List items in every transaction in value descending order R:  $\langle a, f, g, d, b, h, c, e \rangle$ 
  - C is convertible anti-monotone w.r.t. R
- Scan TDB once
  - remove infrequent items
    - Item h is dropped
  - Itemsets a and f are good, ...
- Projection-based mining
  - Imposing an appropriate order on item projection
  - Many tough constraints can be converted into (anti)-monotone

| Item | Value |
|------|-------|
| a    | 40    |
| f    | 30    |
| g    | 20    |
| d    | 10    |
| b    | 0     |
| h    | -10   |
| c    | -20   |
| e    | -30   |

TDB ( $\text{min\_sup}=2$ )

| TID | Transaction                 |
|-----|-----------------------------|
| 10  | a, f, d, b, c               |
| 20  | f, g, d, b, c               |
| 30  | a, f, d, c, e               |
| 40  | f, g, h, c, e <sub>35</sub> |

# Handling Multiple Constraints

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- Different constraints may require different or even conflicting item-ordering
- If there exists an order  $R$  s.t. both  $C_1$  and  $C_2$  are convertible w.r.t.  $R$ , then there is no conflict between the two convertible constraints
- If there exists conflict on order of items
  - Try to satisfy one constraint first
  - Then using the order for the other constraint to mine frequent itemsets in the corresponding projected database

# What Constraints Are Convertible?

| Constraint                                                        | Convertible anti-monotone | Convertible monotone | Strongly convertible |
|-------------------------------------------------------------------|---------------------------|----------------------|----------------------|
| $\text{avg}(S) \leq, \geq v$                                      | Yes                       | Yes                  | Yes                  |
| $\text{median}(S) \leq, \geq v$                                   | Yes                       | Yes                  | Yes                  |
| $\text{sum}(S) \leq v$ (items could be of any value, $v \geq 0$ ) | Yes                       | No                   | No                   |
| $\text{sum}(S) \leq v$ (items could be of any value, $v \leq 0$ ) | No                        | Yes                  | No                   |
| $\text{sum}(S) \geq v$ (items could be of any value, $v \geq 0$ ) | No                        | Yes                  | No                   |
| $\text{sum}(S) \geq v$ (items could be of any value, $v \leq 0$ ) | Yes                       | No                   | No                   |
| .....                                                             |                           |                      |                      |



# Constraint-Based Mining — A General Picture

| Constraint                                             | Anti-monotone | Monotone    | Succinct |
|--------------------------------------------------------|---------------|-------------|----------|
| $v \in S$                                              | no            | yes         | yes      |
| $S \supseteq V$                                        | no            | yes         | yes      |
| $S \subseteq V$                                        | yes           | no          | yes      |
| $\min(S) \leq v$                                       | no            | yes         | yes      |
| $\min(S) \geq v$                                       | yes           | no          | yes      |
| $\max(S) \leq v$                                       | yes           | no          | yes      |
| $\max(S) \geq v$                                       | no            | yes         | yes      |
| $\text{count}(S) \leq v$                               | yes           | no          | weakly   |
| $\text{count}(S) \geq v$                               | no            | yes         | weakly   |
| $\text{sum}(S) \leq v \ (a \in S, a \geq 0)$           | yes           | no          | no       |
| $\text{sum}(S) \geq v \ (a \in S, a \geq 0)$           | no            | yes         | no       |
| $\text{range}(S) \leq v$                               | yes           | no          | no       |
| $\text{range}(S) \geq v$                               | no            | yes         | no       |
| $\text{avg}(S) \theta v, \theta \in \{=, \leq, \geq\}$ | convertible   | convertible | no       |
| $\text{support}(S) \geq \xi$                           | yes           | no          | no       |
| $\text{support}(S) \leq \xi$                           | no            | yes         | no       |

# Chapter 7 : Advanced Frequent Pattern Mining

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- ⑩ Pattern Mining: A Road Map
- ⑩ Pattern Mining in Multi-Level, Multi-Dimensional Space
- ⑩ Constraint-Based Frequent Pattern Mining
- ⑩ Mining High-Dimensional Data and Colossal Patterns 
- ⑩ Mining Compressed or Approximate Patterns
- ⑩ Pattern Exploration and Application
- ⑩ Summary

# Mining Colossal Frequent Patterns

- F. Zhu, X. Yan, J. Han, P. S. Yu, and H. Cheng, “Mining Colossal Frequent Patterns by Core Pattern Fusion”, ICDE'07.
- We have many algorithms, but can we mine large (i.e., colossal) patterns? — such as just size around 50 to 100? Unfortunately, not!
- Why not? — the curse of “downward closure” of frequent patterns
  - The “downward closure” property
    - Any sub-pattern of a frequent pattern is frequent.
  - Example. If  $(a_1, a_2, \dots, a_{100})$  is frequent, then  $a_1, a_2, \dots, a_{100}, (a_1, a_2), (a_1, a_3), \dots, (a_1, a_{100}), (a_1, a_2, a_3), \dots$  are all frequent! There are about  $2^{100}$  such frequent itemsets!
  - No matter using breadth-first search (e.g., Apriori) or depth-first search (FPgrowth), we have to examine so many patterns
- Thus the downward closure property leads to explosion!

# Colossal Patterns: A Motivating Example

**Let's make a set of 40 transactions**

T1 = 1 2 3 4 ... 39

40

T2 = 1 2 3 4 ... 39

40

:

.

:

.

:

**Then delete the items on the diagonal**

:

T<sub>1</sub> = 2 3 4 ... 39 40

T<sub>2</sub> = 1 3 4 ... 39 40

:

.

:

.

:

.

:

.

Closed/maximal patterns may partially alleviate the problem but not really solve it: We often need to mine scattered large patterns!

Let the minimum support threshold  $\sigma = 20$

There are  $\binom{40}{20}$  frequent patterns of size 20

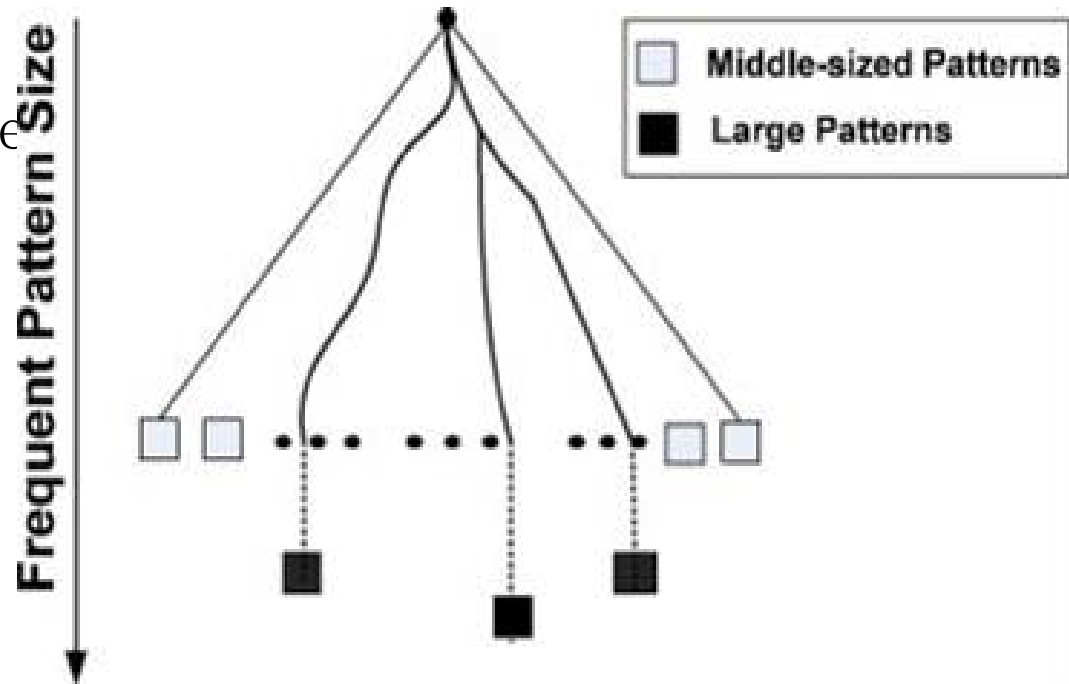
Each is closed and maximal

$$\# \text{ patterns} = \binom{n}{n/2} \approx \sqrt{2/\pi} \frac{2^n}{\sqrt{n}}$$

The size of the answer set is exponential to  $n$

# Colossal Pattern Set: Small but Interesting

- It is often the case that only a small number of patterns are colossal, i.e., of large size
- Colossal patterns are usually attached with greater importance than those of small pattern sizes



# Mining Colossal Patterns: Motivation and Philosophy

---

- Motivation: Many real-world tasks need mining colossal patterns
  - Micro-array analysis in bioinformatics (when support is low)
  - Biological sequence patterns
  - Biological/sociological/information graph pattern mining
- *No hope for completeness*
  - If the mining of mid-sized patterns is explosive in size, there is no hope to find colossal patterns efficiently by insisting “complete set” mining philosophy
- *Jumping out of the swamp of the mid-sized results*
  - What we may develop is a philosophy that may jump out of the swamp of mid-sized results that are explosive in size and jump to reach colossal patterns
- *Striving for mining almost complete colossal patterns*
  - The key is to develop a mechanism that may quickly reach

# Alas, A Show of Colossal Pattern Mining!

$T_1 = 2 \ 3 \ 4 \ \dots \ 39 \ 40$   
 $T_2 = 1 \ 3 \ 4 \ \dots \ 39 \ 40$   
 $:$   
 $:$   
 $:$   
 $:$   
 $:$   
 $:$   
 $:$   
 $T_{41} = 1 \ 2 \ 3 \ 4 \ \dots \ 79$   
 $T_{42} = 41 \ 42 \ 43 \ \dots \ 79$   
 $:$   
 $:$   
 $T_{60} = 41 \ 42 \ 43 \ \dots \ 79$

Let the min-support threshold  $\sigma = 20$

Then there are  $\binom{40}{20}$  closed/maximal frequent patterns of size 20

However, there is only one with size greater than 20, (*i.e.*, colossal):

$\alpha = \{41, 42, \dots, 79\}$  of size 39

The existing fastest mining algorithms (*e.g.*, FPClose, LCM) fail to complete running

Our algorithm outputs this colossal pattern in seconds

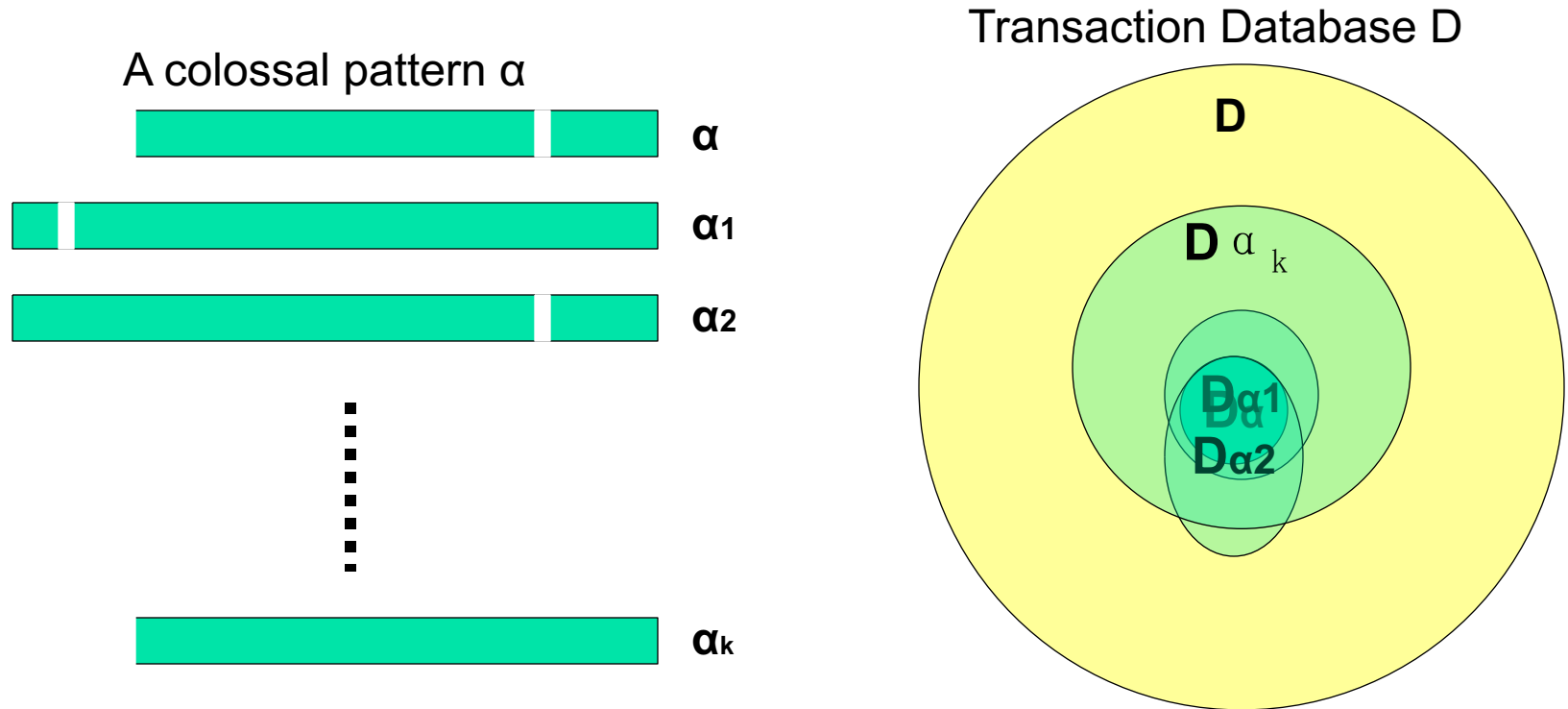
# Methodology of Pattern-Fusion Strategy

---

- Pattern-Fusion traverses the tree in a bounded-breadth way
  - Always pushes down a frontier of a bounded-size candidate pool
  - Only a fixed number of patterns in the current candidate pool will be used as the starting nodes to go down in the pattern tree — thus avoids the exponential search space
- Pattern-Fusion identifies “shortcuts” whenever possible
  - Pattern growth is not performed by single-item addition but by leaps and bounded: agglomeration of multiple patterns in the pool
  - These shortcuts will direct the search down the tree much more rapidly towards the colossal patterns



# Observation: Colossal Patterns and Core Patterns



Subpatterns  $\alpha_1$  to  $\alpha_k$  cluster tightly around the colossal pattern  $\alpha$  by sharing a similar support. We call such subpatterns *core patterns* of  $\alpha$

# Robustness of Colossal Patterns

---

- Core Patterns

Intuitively, for a frequent pattern  $\alpha$ , a subpattern  $\beta$  is a  $\tau$ -core pattern of  $\alpha$  if  $\beta$  shares a similar support set with  $\alpha$ , i.e.,

$$\frac{|D_\alpha|}{|D_\beta|} \geq \tau \quad 0 < \tau \leq 1$$

where  $\tau$  is called the core ratio

- Robustness of Colossal Patterns

A colossal pattern is robust in the sense that it tends to have much more core patterns than small patterns

# Example: Core Patterns

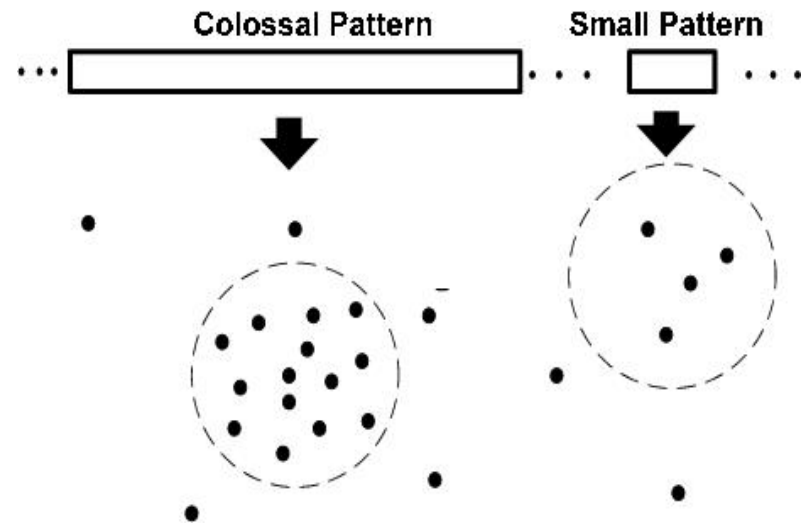
- A colossal pattern has far more core patterns than a small-sized pattern
- A colossal pattern has far more core descendants of a smaller size  $c$
- A random draw from a complete set of pattern of size  $c$  would more likely to pick a core descendant of a colossal pattern

| Transaction (# of Ts) | Core Patterns ( $\tau = 0.5$ )                                                                                                                                                  |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (abe) (100)           | (abe), (ab), (be), (ae), (e)                                                                                                                                                    |
| (bcf) (100)           | (bcf), (bc), (bf)                                                                                                                                                               |
| (acf) (100)           | (acf), (ac), (af)                                                                                                                                                               |
| (abcef) (100)         | (ab), (ac), (af), (ae), (bc), (bf), (be) (ce), (fe), (e), (abc), (abf), (abe), (ace), (acf), (afe), (bcf), (bce), (bfe), (cfe), (abcf), (abce), (bcfe), (acfe), (abfe), (abcef) |

# Colossal Patterns Correspond to Dense Balls

---

- Due to their robustness, colossal patterns correspond to dense balls
  - $\Omega(2^d)$  in population
- A random draw in the pattern space will hit somewhere in the ball with high probability



# Idea of Pattern-Fusion Algorithm

---

- Generate a complete set of frequent patterns up to a small size
- Randomly pick a pattern  $\beta$ , and  $\beta$  has a high probability to be a core-descendant of some colossal pattern  $\alpha$
- Identify all  $\alpha$ 's descendants in this complete set, and merge all of them — This would generate a much larger core-descendant of  $\alpha$
- In the same fashion, we select  $K$  patterns. This set of larger core-descendants will be the candidate pool for the next iteration

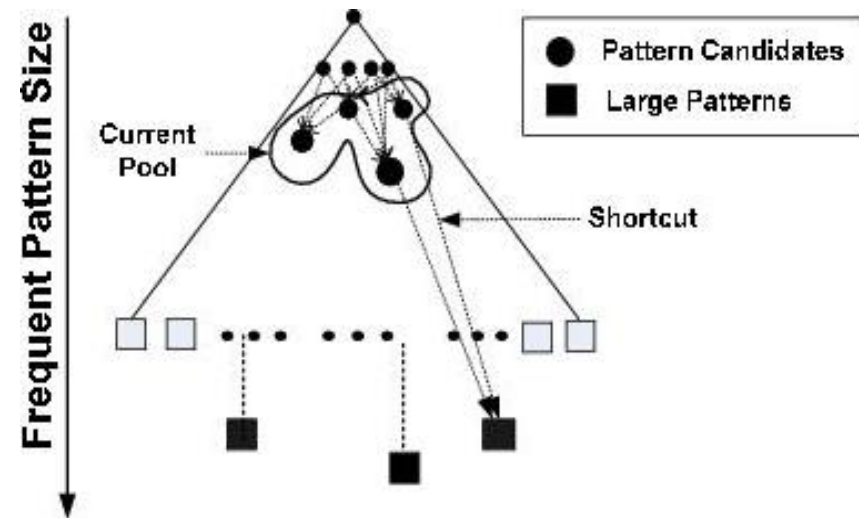
# Pattern-Fusion: The Algorithm

---

- Initialization (Initial pool): Use an existing algorithm to mine all frequent patterns up to a small size, e.g., 3
- Iteration (Iterative Pattern Fusion):
  - At each iteration,  $k$  seed patterns are randomly picked from the current pattern pool
  - For each seed pattern thus picked, we find all the patterns within a bounding ball centered at the seed pattern
  - All these patterns found are fused together to generate a set of super-patterns. All the super-patterns thus generated form a new pool for the next iteration
- Termination: when the current pool contains no more

# Why Is Pattern-Fusion Efficient?

- A bounded-breadth pattern tree traversal
  - It avoids explosion in mining mid-sized ones
  - Randomness comes to help to stay on the right path
- Ability to identify “short-cuts” and take “leaps”
  - fuse small patterns together in one step to generate new patterns of significant sizes



# Pattern-Fusion Leads to Good Approximation

---

- Gearing toward colossal patterns
  - The larger the pattern, the greater the chance it will be generated
- Catching outliers
  - The more distinct the pattern, the greater the chance it will be generated



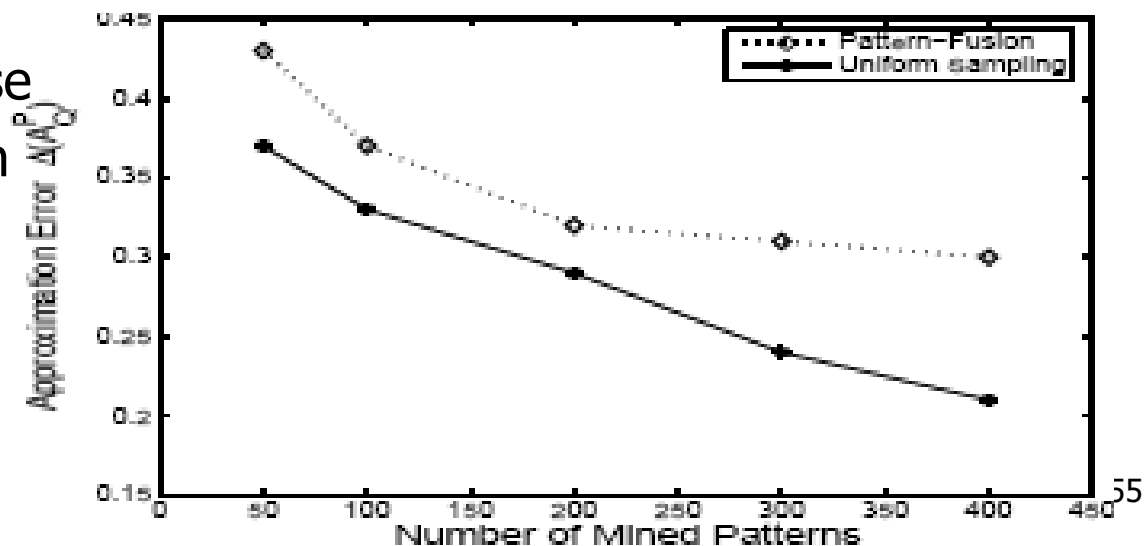
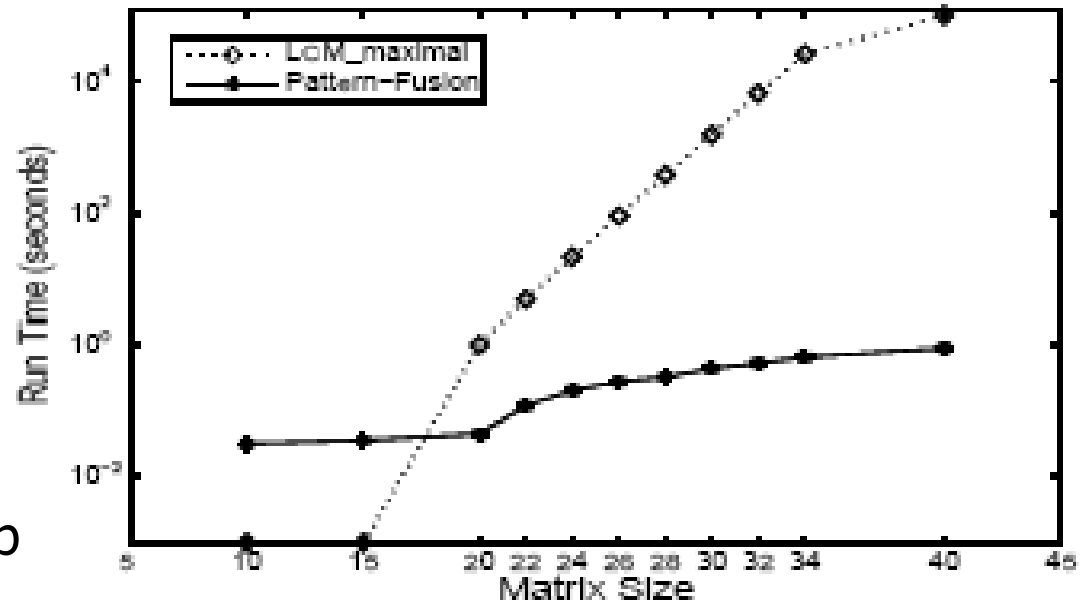
# Experimental Setting

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- Synthetic data set
  - $\text{Diag}_n$  an  $n \times (n-1)$  table where  $i^{\text{th}}$  row has integers from 1 to  $n$  except  $i$ . Each row is taken as an itemset.  $\text{min\_support}$  is  $n/2$ .
- Real data set
  - Replace: A program trace data set collected from the “replace” program, widely used in software engineering research
  - ALL: A popular gene expression data set, a clinical data on ALL-AML leukemia ([www.broad.mit.edu/tools/data.html](http://www.broad.mit.edu/tools/data.html)).
    - Each item is a column, representing the activity level of gene/protein in the same
    - Frequent pattern would reveal important correlation between gene expression patterns and disease outcomes

# Experiment Results on $\text{Diag}_n$

- LCM run time increases exponentially with pattern size  $n$
- Pattern-Fusion finishes efficiently
- The approximation error of Pattern-Fusion (with min-sup 20) in comparison with the complete set) is rather close to uniform sampling (which randomly picks  $K$  patterns from the complete answer set)



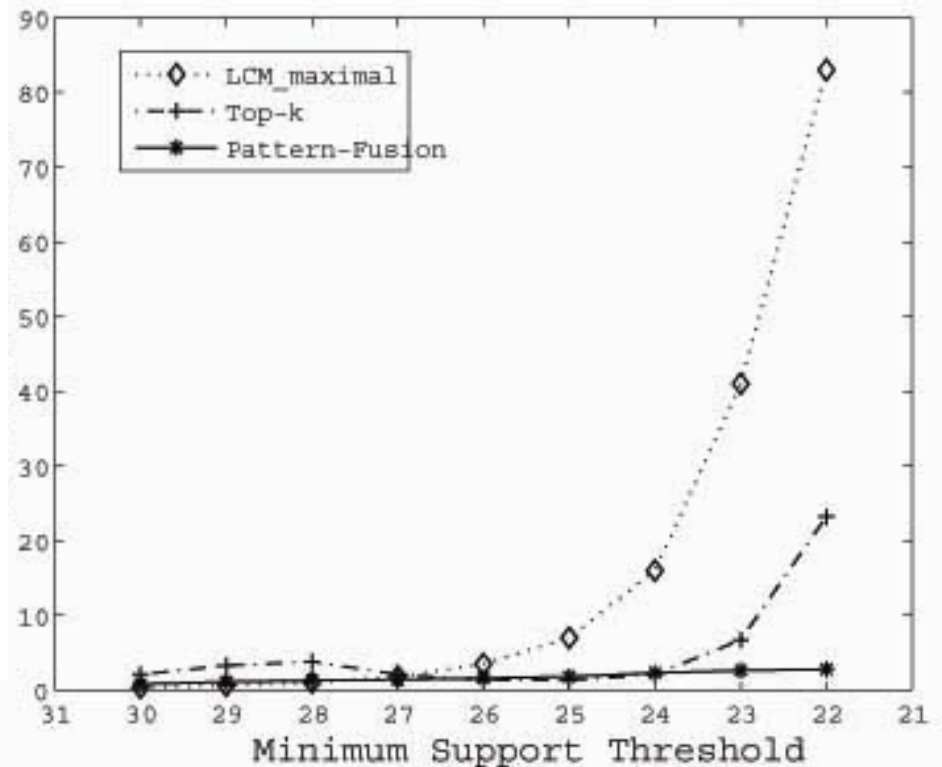
# Experimental Results on ALL

- ALL: A popular gene expression data set with 38 transactions, each with 866 columns
  - There are 1736 items in total
  - The table shows a high frequency threshold of

|                  |     |     |     |    |    |    |    |
|------------------|-----|-----|-----|----|----|----|----|
| Pattern Size     | 110 | 107 | 102 | 91 | 86 | 84 | 83 |
| The complete set | 1   | 1   | 1   | 1  | 1  | 2  | 6  |
| Pattern-Fusion   | 1   | 1   | 1   | 1  | 1  | 1  | 4  |

|                  |    |    |    |    |    |    |    |
|------------------|----|----|----|----|----|----|----|
| Pattern Size     | 82 | 77 | 76 | 75 | 74 | 73 | 71 |
| The complete set | 1  | 2  | 1  | 1  | 1  | 2  | 1  |
| Pattern-Fusion   | 0  | 2  | 0  | 1  | 1  | 1  | 1  |



# Experimental Results on REPLACE

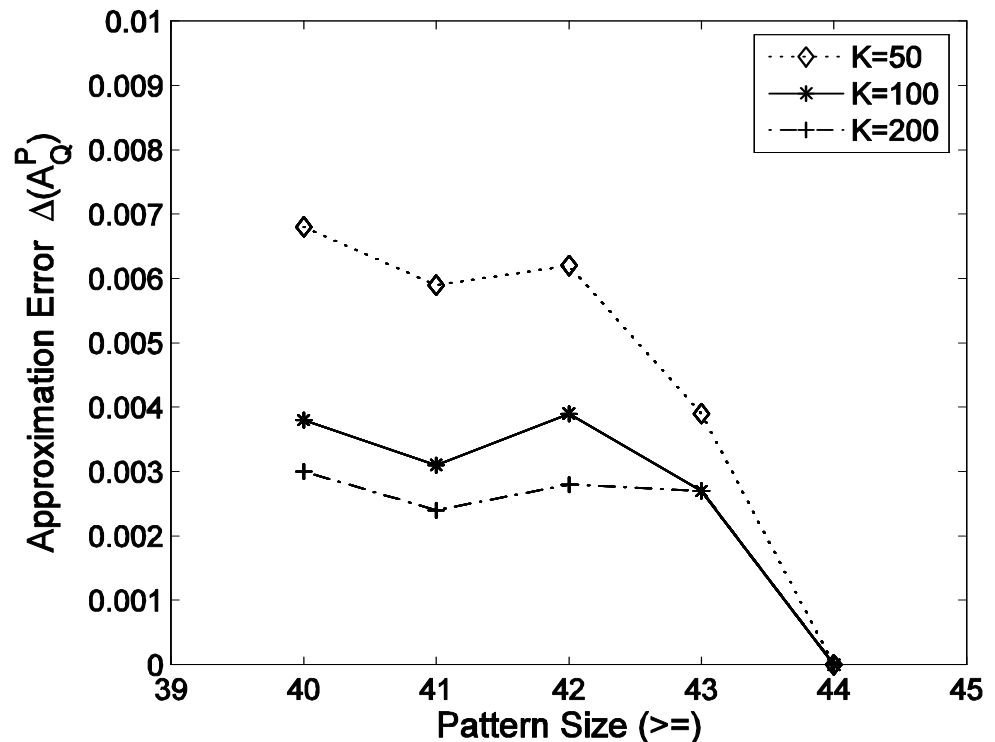
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## ■ REPLACE

- A program trace data set, recording 4395 calls and transitions
- The data set contains 4395 transactions with 57 items in total
- With support threshold of 0.03, the largest patterns are of size 44
- They are all discovered by Pattern-Fusion with different settings of  $K$  and  $\tau$ , when started with an initial pool of 20948 patterns of size  $\leq 3$

# Experimental Results on REPLACE

- Approximation error when compared with the complete mining result
- Example. Out of the total 98 patterns of size  $\geq 42$ , when  $K=100$ , Pattern-Fusion returns 80 of them
- A good approximation to the colossal patterns in the sense that any pattern in the complete set is on average at most 0.17 items away from one of these 80 patterns



# Chapter 7 : Advanced Frequent Pattern Mining

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# Mining Compressed Patterns: $\delta$ -clustering

- Why compressed patterns?

- too many, but less meaningful

- Pattern distance: 
$$D(P_1, P_2) = 1 - \frac{|T(P_1) \cap T(P_2)|}{|T(P_1) \cup T(P_2)|}$$

- $\delta$ -clustering: For each pattern P, find all patterns which can be expressed by P and their distance to P are within  $\delta$  ( $\delta$ -cover)

- All patterns in the cluster can be represented by P

- Xin et al., "Mining Compressed Frequent-Pattern Sets" VLDB'05

| ID | Item-Sets                | Support |
|----|--------------------------|---------|
| P1 | {38, 16, 18, 12}         | 205227  |
| P2 | {38, 16, 18, 12, 17}     | 205211  |
| P3 | {39, 38, 16, 18, 12, 17} | 101758  |
| P4 | {39, 16, 18, 12, 17}     | 161563  |
| P5 | {39, 16, 18, 12}         | 161576  |

- Closed frequent pattern

- Report P1, P2, P3, P4, P5
  - Emphasize too much on support
  - no compression

- Max-pattern, P3: info loss

- A desirable output: P2, P3, P4

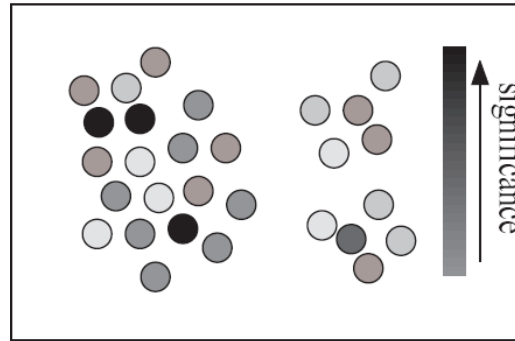
# Redundancy-Award Top-k Patterns

- Why redundancy-aware top-k patterns?

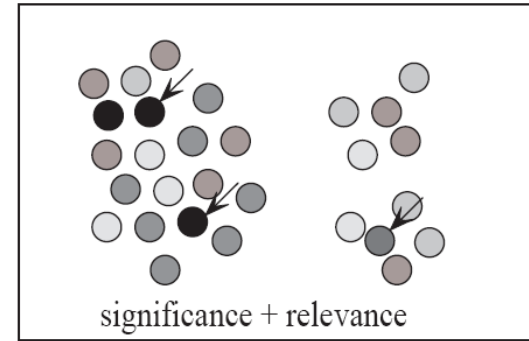
- Desired patterns: high significance & low redundancy

- Propose the MMS (Maximal Marginal Significance) for measuring the combined significance of a pattern set

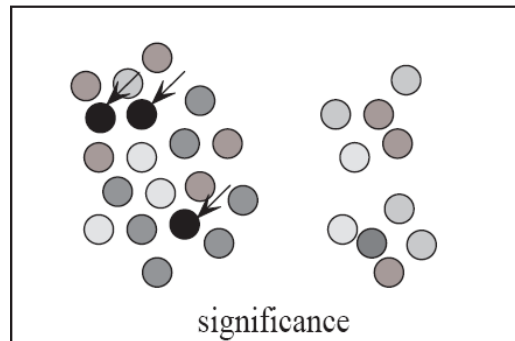
- Xin et al., Extracting Redundancy-Aware Top-K Patterns, KDD'06



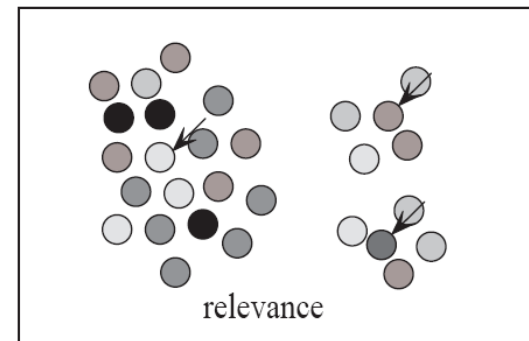
(a) a set of patterns



(b) redundancy-aware top-k



(c) traditional top-k



(d) summarization



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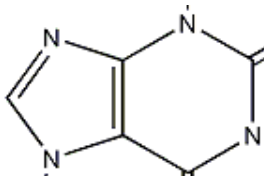
# How to Understand and Interpret Patterns?

diaper

beer

- Do they all make sense?
- What do they mean?
- How are they useful?

female sterile (2) tekele



morphological info. and simple statistics



Semantic Information

*Not all frequent patterns are useful, only meaningful ones ...*



Annotate patterns with semantic information

# A Dictionary Analogy

## Word: "pattern" – from Merriam-Webster

Main Entry: **1** *pat-tern*

Pronunciation: 'pa-tern

Function: *noun*

Etymology: Middle English *patron*, from Middle French, from Latin *patronus*

Date: 14th century

Non-semantic info.

Definitions indicating semantics

1 : a form

2 : something

*pattern*

3 : a model

4 : an artist

5 : a nature

Main Entry:

**pattern**

Function:

*noun*

Synonyms

**nonynms**

MODEL 2, archetype, beau ideal, ensample, example, exemplar, ideal,

error, paradigm, standard

component

**Related Word**

*original*

12 : freque

2

**synonyms**

**Synonyms**

FIGURE 3, design, device, motif, motive

-*pat-tern*

**Related Word**

*patterning*

Related Words

**nonynms**

ORDER 8, method, orderliness, plan, system

**Related Word**

*arrangement, constellation*

# Semantic Analysis with Context Models

---

- Task1: Model the context of a frequent pattern

*Based on the Context Model...*

- Task2: Extract strongest context indicators
- Task3: Extract representative transactions
- Task4: Extract semantically similar patterns

# Annotating DBLP Co-authorship & Title Pattern

Database:

| Authors               | Title                                                |
|-----------------------|------------------------------------------------------|
| X. Yan, P. Yu, J. Han | Substructure Similarity Search<br>in Graph Databases |
| ...                   | ...                                                  |
| ...                   | ...                                                  |

Frequent Patterns

$P_1: \{x\_yan, j\_han\}$

Frequent

$P_2: \text{“substructure search”}$

Semantic Annotations

| Pattern | { x_yan, j_han }               |
|---------|--------------------------------|
| Non     | Sup = ...                      |
| CI      | {p_yu}, graph pattern, ...     |
| Trans.  | <b>gSpan: graph-base.....</b>  |
| SSPs    | { j_wang }, {j_han, p_yu}, ... |

Context Units

$\langle \{ p\_yu, j\_han \}, \{ d\_xin \}, \dots, \text{“graph pattern”}, \dots, \text{“substructure similarity”}, \dots \rangle$

Pattern = {xifeng\_yan, jiawei\_han}

Annotation Results:

|                                     |                                                                                                             |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Context Indicator (CI)              | graph; {philip_yu}; mine close; graph pattern; sequential pattern; ...                                      |
| Representative Transactions (Trans) | > gSpan: graph-base substructure pattern mining;<br>> mining close relational graph connect constraint; ... |
| Semantically Similar Patterns (SSP) | {jiawei_han, philip_yu}; {jian_pei, jiawei_han}; {jiong_yang, philip_yu, wei_wang}; ...                     |

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# Summary

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- ⑩ Roadmap: Many aspects & extensions on pattern mining
- ⑩ Mining patterns in multi-level, multi dimensional space
- ⑩ Mining rare and negative patterns
- ⑩ Constraint-based pattern mining
- ⑩ Specialized methods for mining high-dimensional data and colossal patterns
- ⑩ Mining compressed or approximate patterns
- ⑩ Pattern exploration and understanding: Semantic annotation of frequent patterns

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---

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- J. Pei, J. Han, and L. V. S. Lakshmanan. Mining Frequent Itemsets with Convertible Constraints. ICDE'01
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- F. Zhu, X. Yan, J. Han, and P. S. Yu, “gPrune: A Constraint Pushing Framework for Graph Pattern Mining”, PAKDD'07

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